Alex Dils

Berkeley, CA | dils@berkeley.edu | alex-dils.com | Scholar | LinkedIn | GitHub

Website

GitHub

Education

University of California, Berkeley Junior
B.A. in ***Computer Science*** | GPA: 3.7/4.0
Relevant Coursework: **CS 61A/B:** programming abstractions, data structures, algorithms, Java/Python; **CS 61C:** C, memory, RISC-V. **CS 70:** discrete math, proofs, graph theory, combinatorics, probability; **Calculus I–III:** single/multivariable and vector calculus.
Extracurriculars: **Machine Learning at Berkeley:** Internal & Research Committees; **Reverb**

Technical Skills

Python, PyTorch, TensorFlow, scikit-learn, OpenCV, SimpleITK, Core ML, ARKit, Metal, VLMs, multimodal learning, medical imaging, model evaluation, prompt engineering, data pipelines, FastAPI, Node/TypeScript, React, SQL/Postgres, Supabase/PostGIS, Docker, Slurm

Experience

Machine Learning Engineer | Logitech Feb 2026 – Present
- Built a multi-model **VLM evaluation harness** for meeting-room readiness with structured prompting/parsing, 28 ablations (7 models $\times$ 4 strategies), and **0% parse failures** across 504 predictions.
- Ran latency/accuracy benchmarking across VLMs and CV baselines (YOLOv11, MobileNet-SSD); found SmolVLM2-2.2B matched 4B–8B models while running **27$\times$ faster**, shaping edge-inference deployment tradeoffs.

Founder & CEO, Infrastruct — Agentic Workflow Automation Platform 2026 – Present
- Built an **AI agent platform** with Electron desktop observer + Node backend that mines repeated workflows, matches automation templates, and promotes high-confidence findings into approval-gated automations.
- Designed agent manifests, connector scopes, evidence packs, rollback/idempotency plans, always-on Codex/Telegram agent loops, tracing, OAuth/SSO, billing, deployment, and SOC 2 readiness workflows.

Canary CREST Research Scholar, Stanford Canary Center / Gevaert Lab — Stanford, CA Summer 2026
- Built a **cNF volume-estimation pipeline** for NF1 cancer-risk surveillance, combining center-coordinate lesion localization, SAM v2 masks, ResNet50 scale recovery, Depth Pro surface reconstruction, and per-lesion cm^3 integration.
- Generated synthetic RGB/depth lesion data on Human Skin Repository 3D surfaces with physics-based protrusions and diffusion inpainting; fine-tuned Depth Pro, reaching **52.6** Abs-rel reduction and $R^2 = 0.97$ scale prediction after ruler inpainting.

CTO & Co-Founder, Appraise AI — Berkeley, CA | appraiseai.co Nov 2025 – Mar 2026
- Led ML roadmap, architecture, and enterprise technical strategy for a resale intelligence startup; raised **$15K** pre-seed.
- Architected **multimodal pricing engine** (computer vision + NLP) with **>95% top-3 accuracy**; built Scrapy/Playwright ingestion collecting **500K+ listings/day**.

Research Intern, Stanford Center for Biomedical Informatics Research — Remote Jun 2022 – Present
- Developed **bias-mitigation** and domain-robustness methods for skin-lesion segmentation; built finite-element lung-motion augmentations that **improved Dice by >0.05** on tumor segmentation.
- Contributed to Digital Twin research **featured on Stanford Medicine's homepage**; productionized medical-imaging workflows with PyTorch, TensorFlow, SimpleITK, Slurm, OpenCV, and scikit-learn.

Leadership

StreetCode Academy: designed and taught Python curriculum to 30+ students across four 8-week cohorts.

Projects

SiteRocket | siterocket.io 2026
- Built agentic workflows to generate **1,000+ websites** for Bay Area small business owners, then sold them through proof packets, targeted outreach, and Stripe/CRM purchase flows.

Openleaf | alex-dils.com/openleaf 2026
- Built an **agent-native local LaTeX workspace** in Electron/Node with CodeMirror editing, PDF.js preview, Tectonic/latexmk/pdflatex compile flows, docked Shell/Codex/Claude/SSH terminals, Vim shortcuts, GitHub sync, and macOS CLI/app-bundle install tooling.

Calshi | calshi.app 2025–2026
- Shipped a React/Node/SQL full-stack product with authentication, database-backed user flows, and production deployment, reaching **100+ active users**.

Selected Publications

- **Enhancing Medical AI with Physiologically-Informed Data Augmentation**, ICML 2026 submission (pending).
- **Eye For An Eye: A Deep-Learning and Analytical Method to Spatializing Stereoscopic Images**, National High School Journal of Science, 2025.
- **Microplastic Identification Using AI-Driven Image Segmentation and GAN-Generated Ecological Context**, arXiv, 2024.
- **Addressing Bias and Confounders in AI-based Image Diagnosis — A Study of 117,610 Skin Lesions**, Cell (in revision), 2025.
- **Integrating Mechanistic Knowledge into Deep Learning for Improved Cancer Detection**, FEniCS Conference Proceedings, 2024.